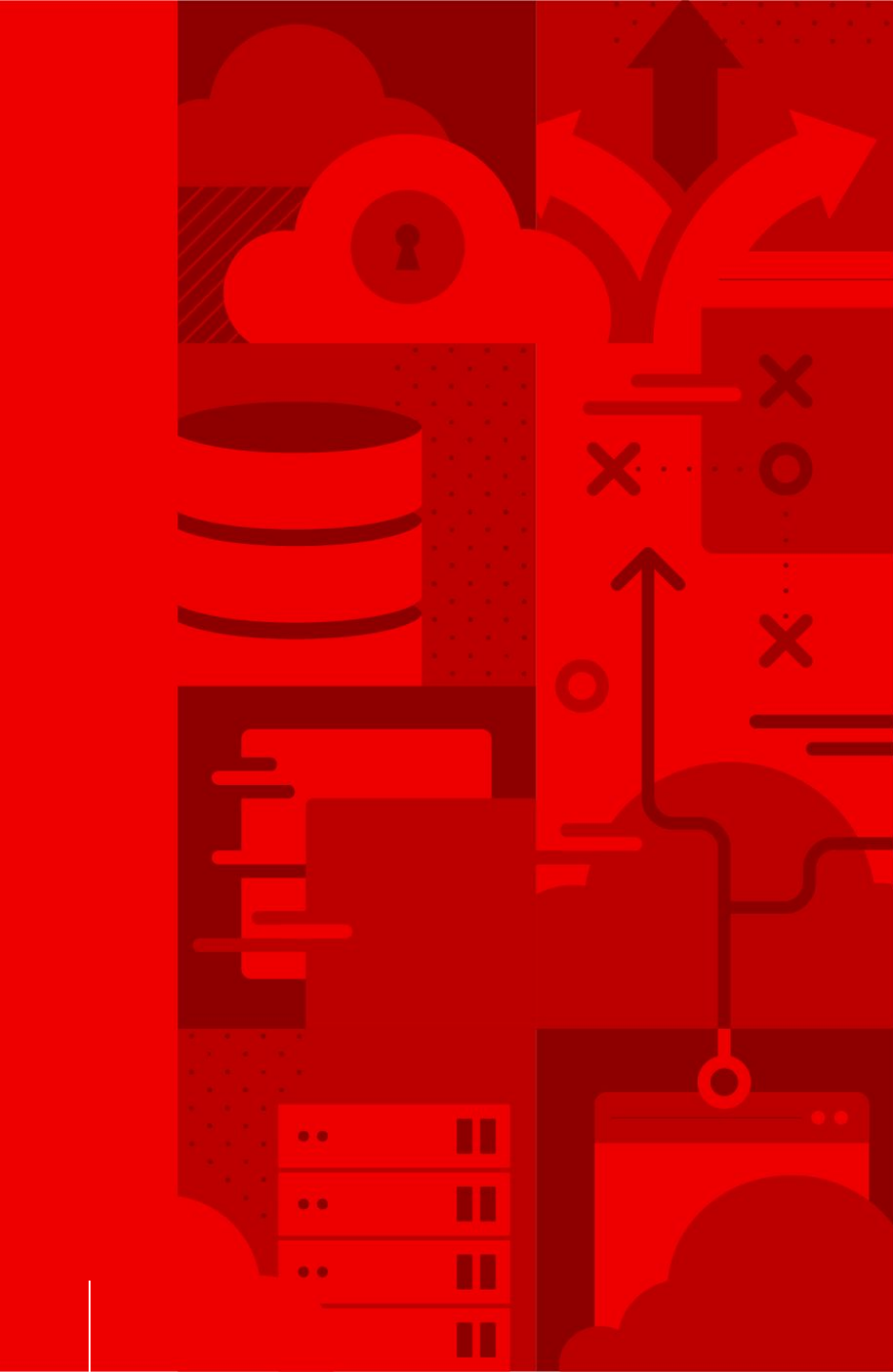




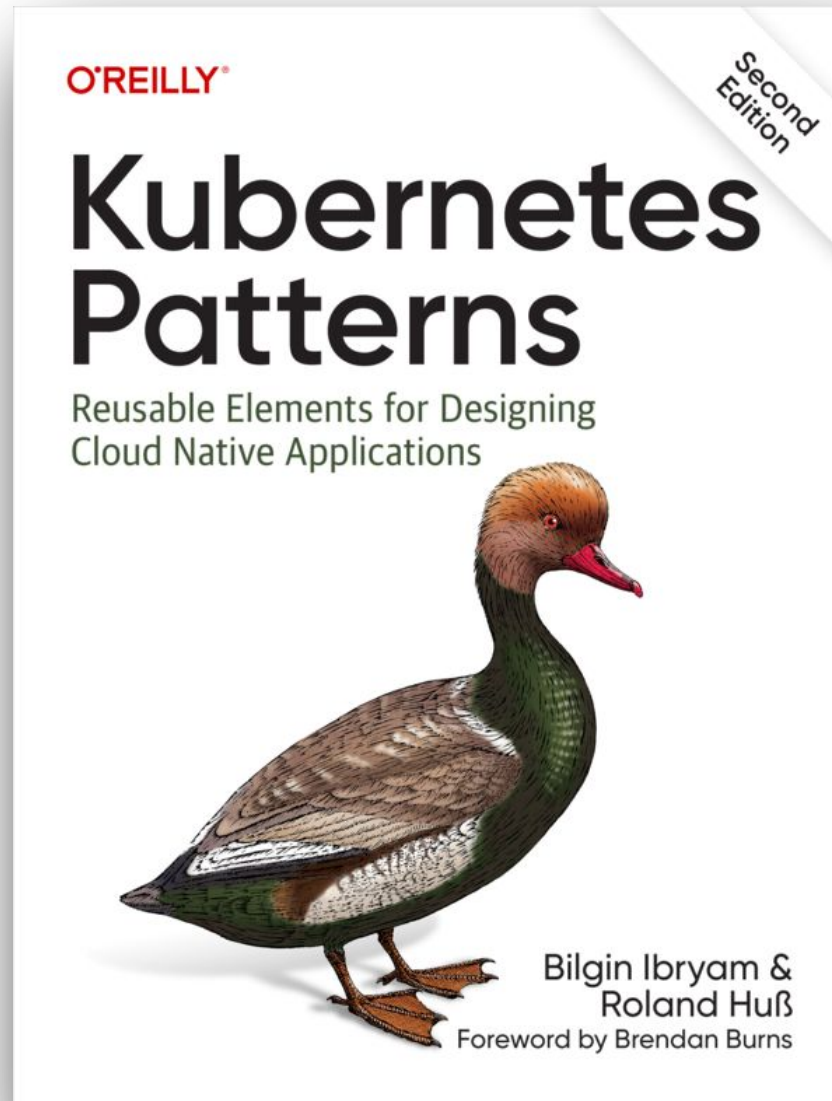
Kubernetes Patterns

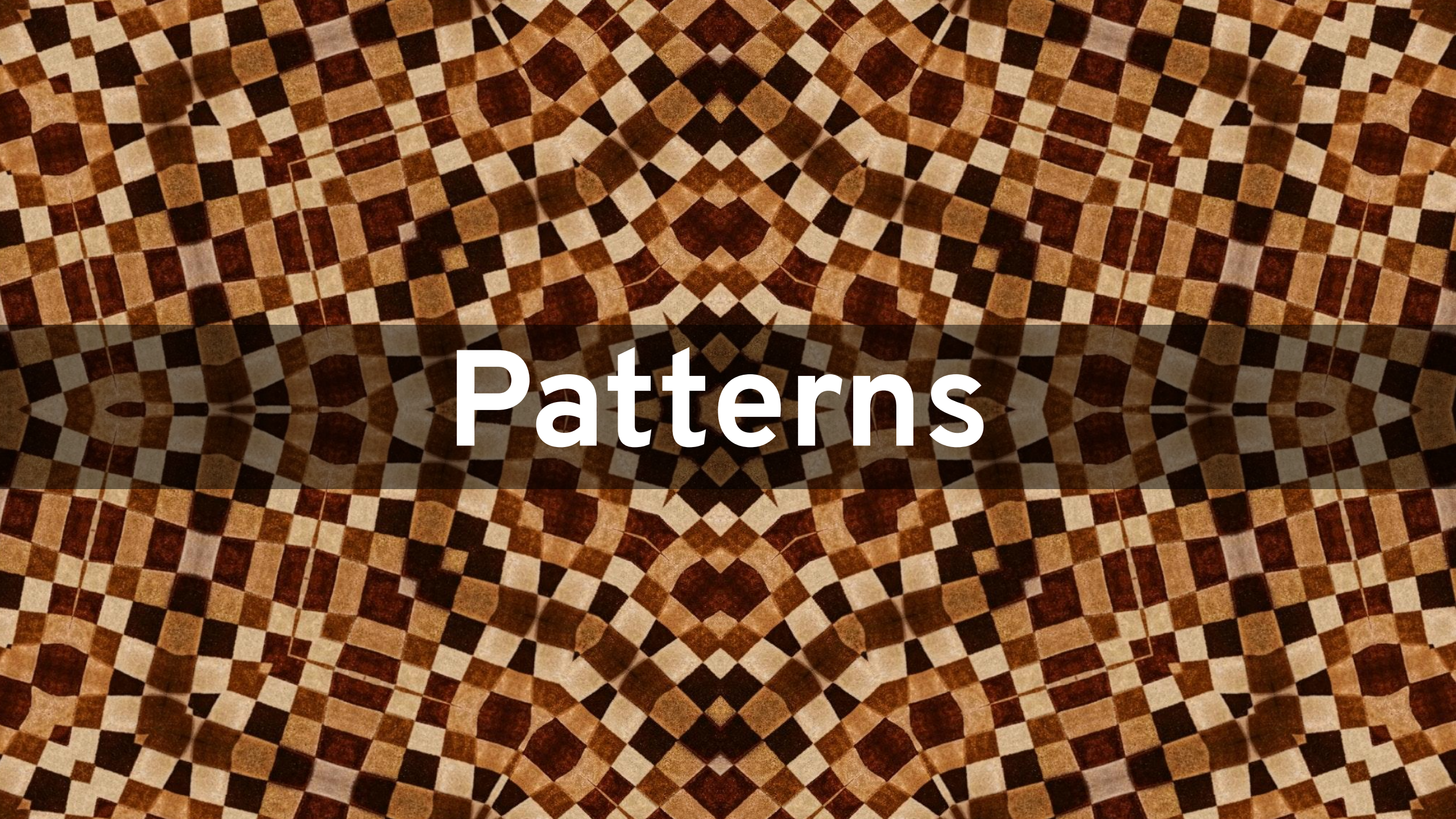
Reusable Elements for Designing
Cloud-Native Applications

Dr. Roland Huß
Distinguished Engineer

@ro14nd@hachyderm.io
www.linkedin.com/in/ro14nd
<https://k8spatterns.com>







Patterns

A Pattern Language

Towns · Buildings · Construction



Christopher Alexander

Sara Ishikawa · Murray Silverstein

WITH

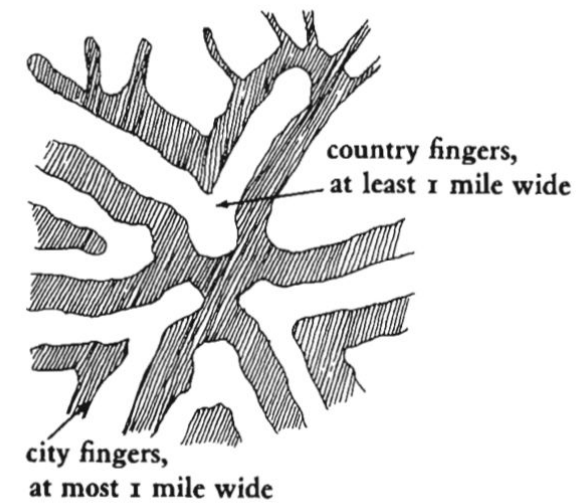
Max Jacobson · Ingrid Fiksdahl-King

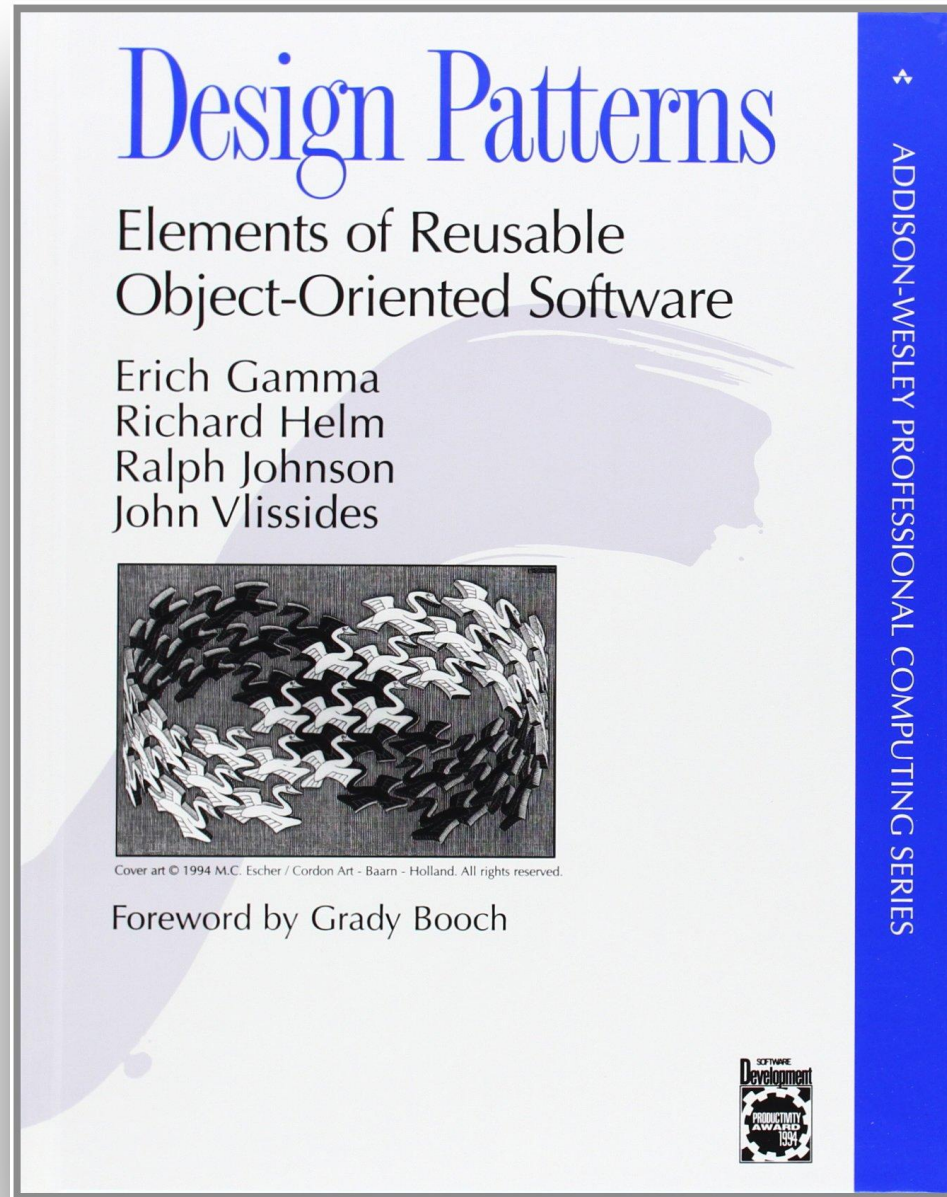
Shlomo Angel

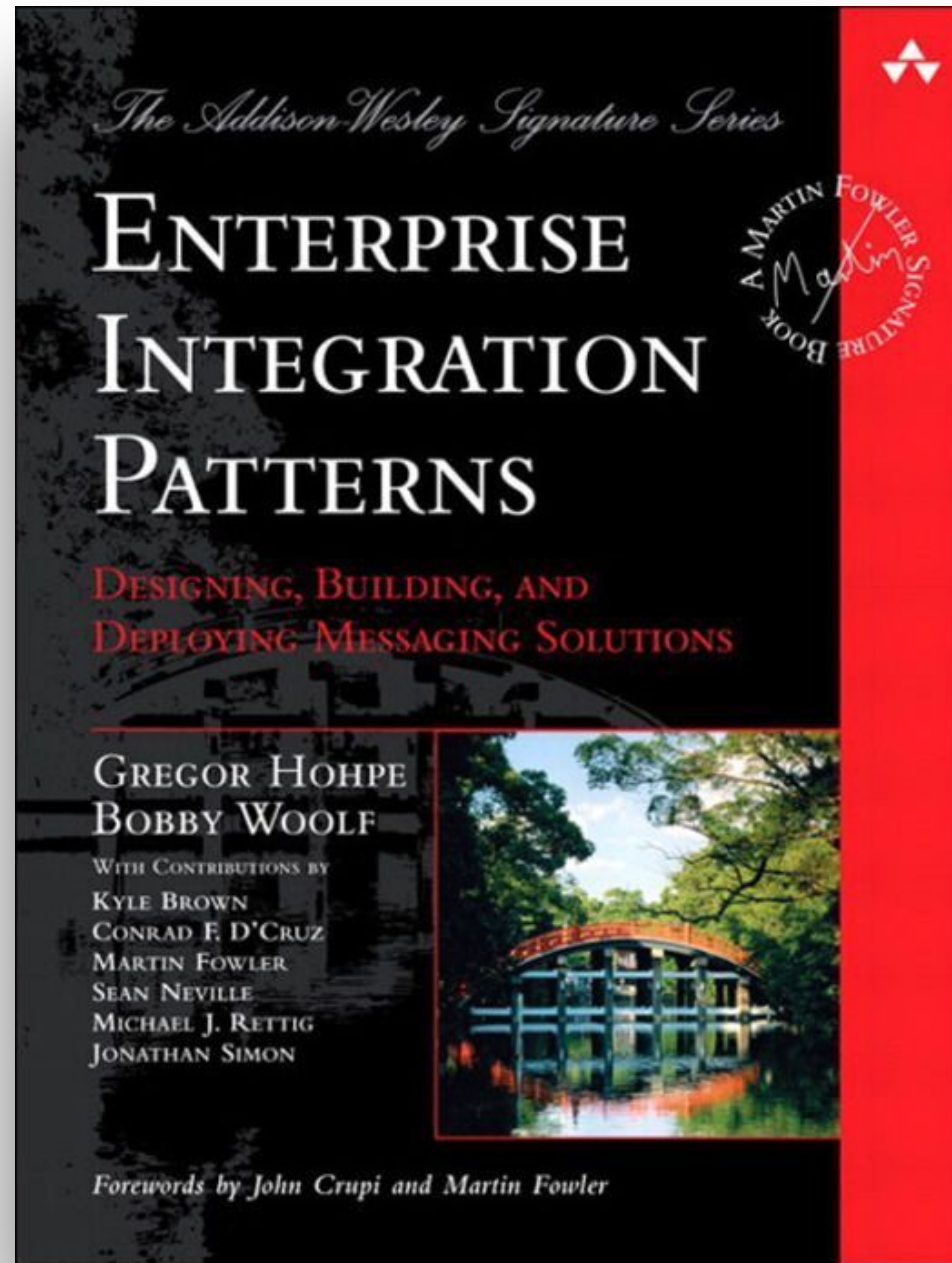
3 CITY COUNTRY FINGERS**



*When the countryside is far away
the city becomes a prison.*









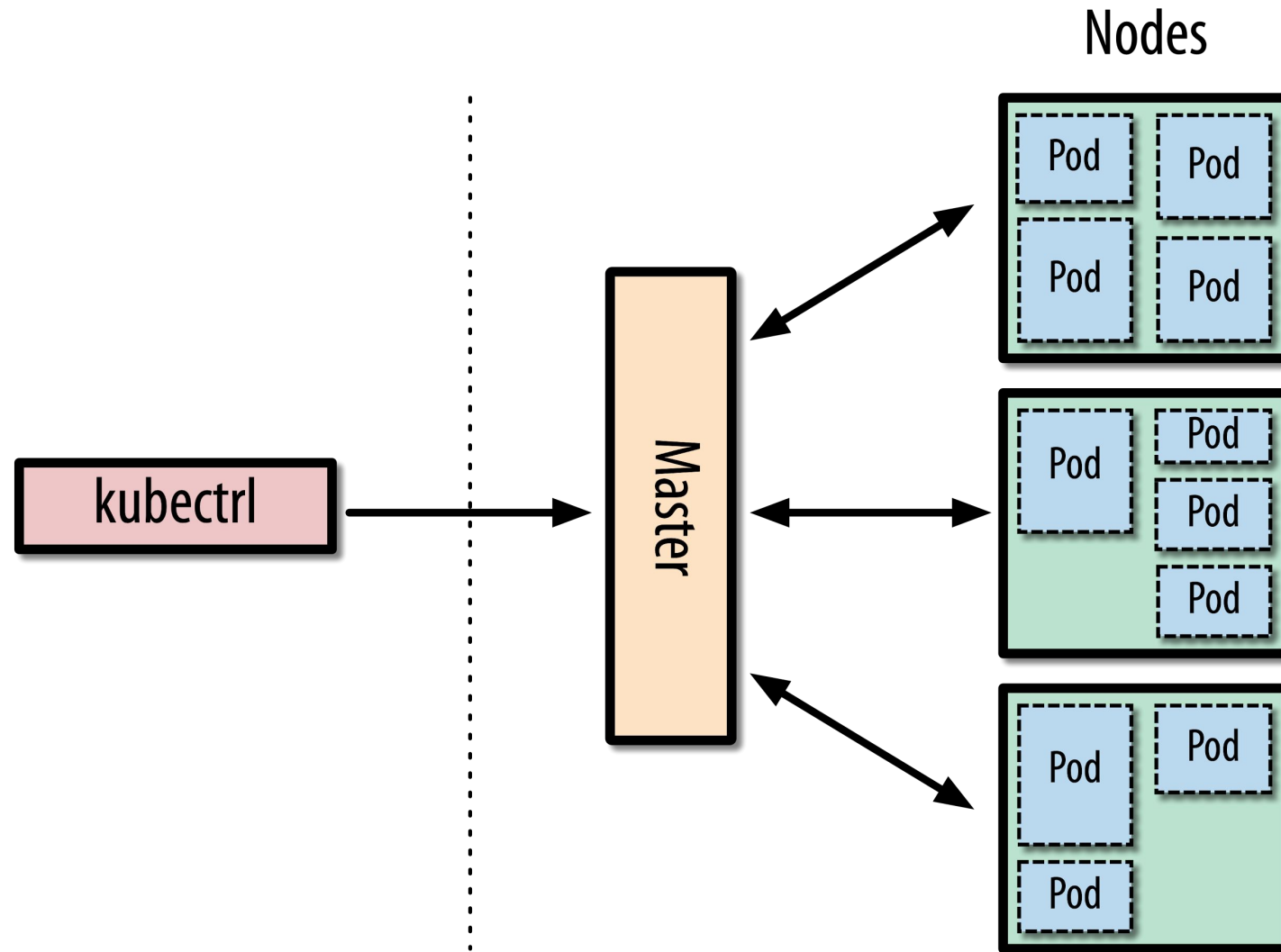
Kubernetes



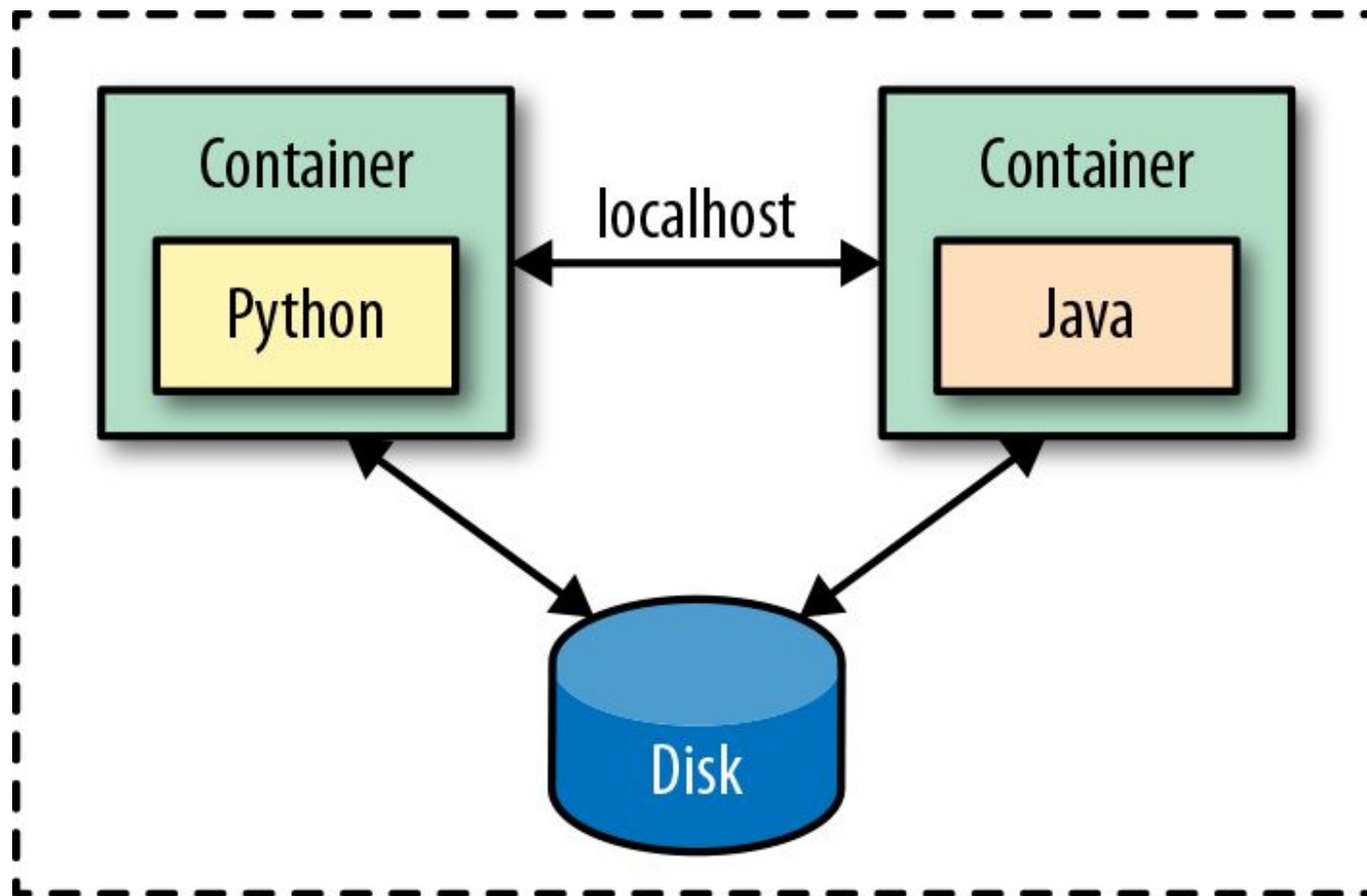
Kubernetes

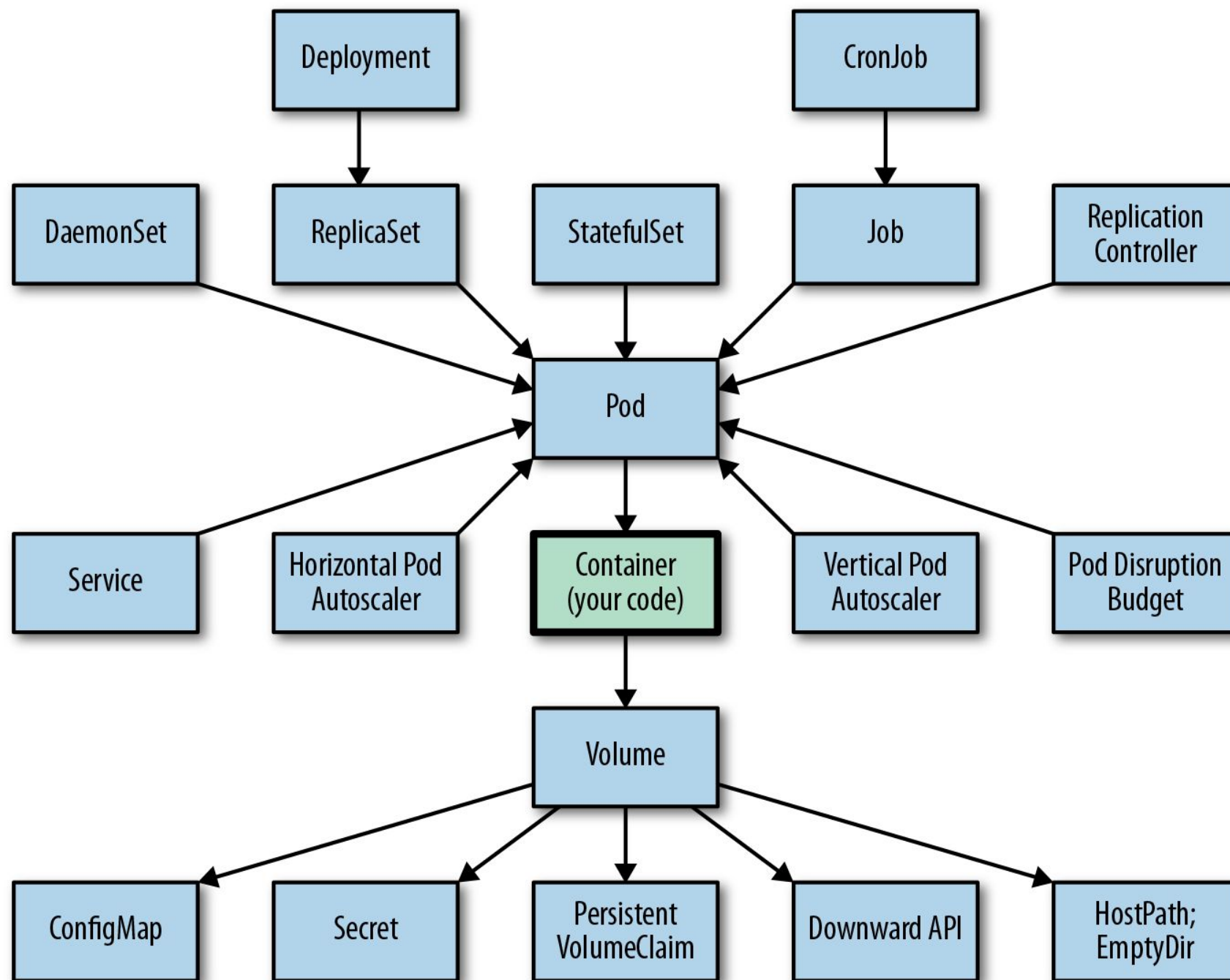
- Open Source container orchestration system started by Google in 2014
 - ✧ Scheduling
 - ✧ Self-healing
 - ✧ Horizontal and vertical scaling
 - ✧ Service discovery
 - ✧ Rollout and Rollbacks
- Declarative resource-centric REST API

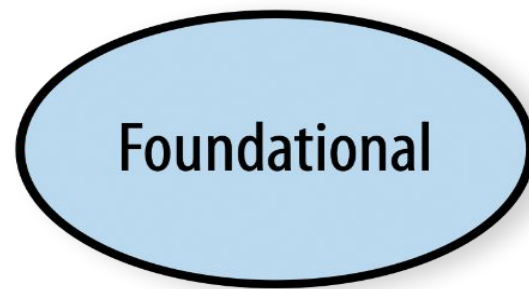
Architecture

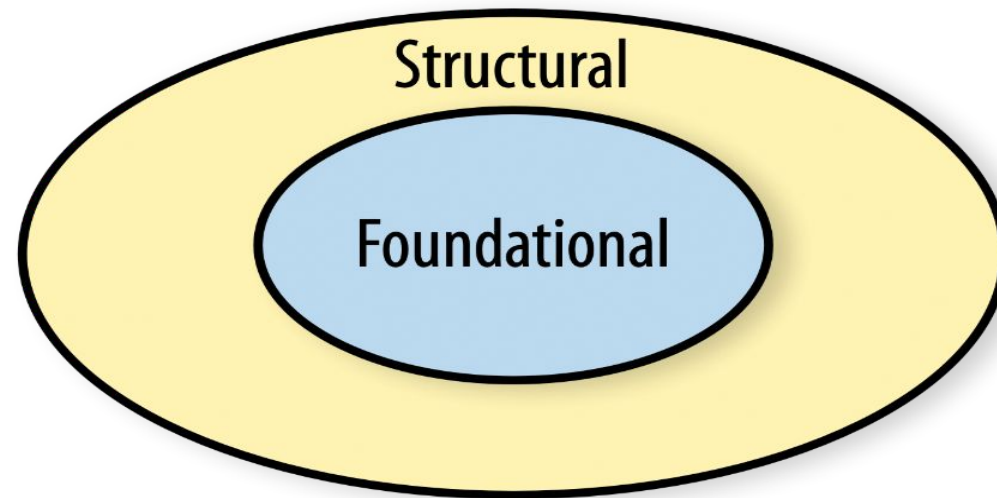


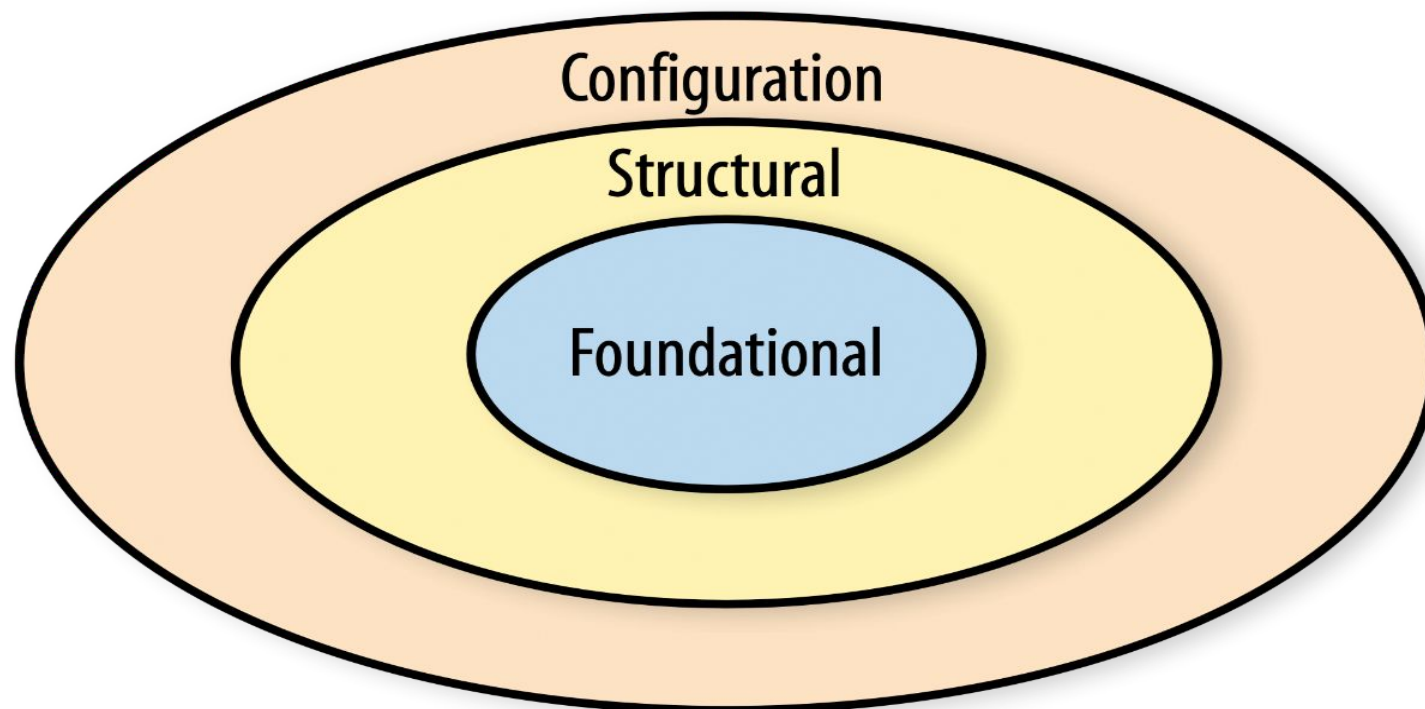
Pod

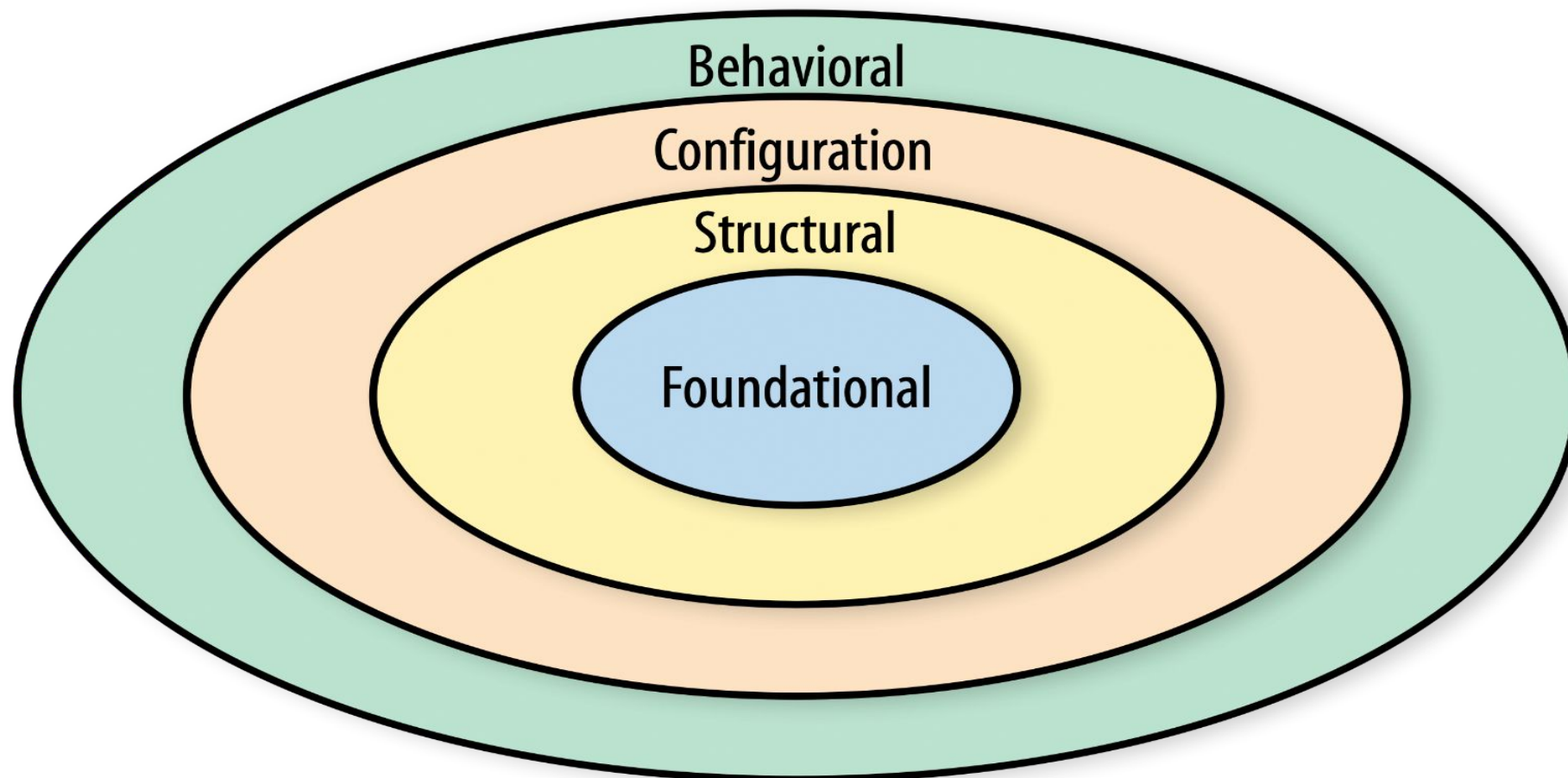


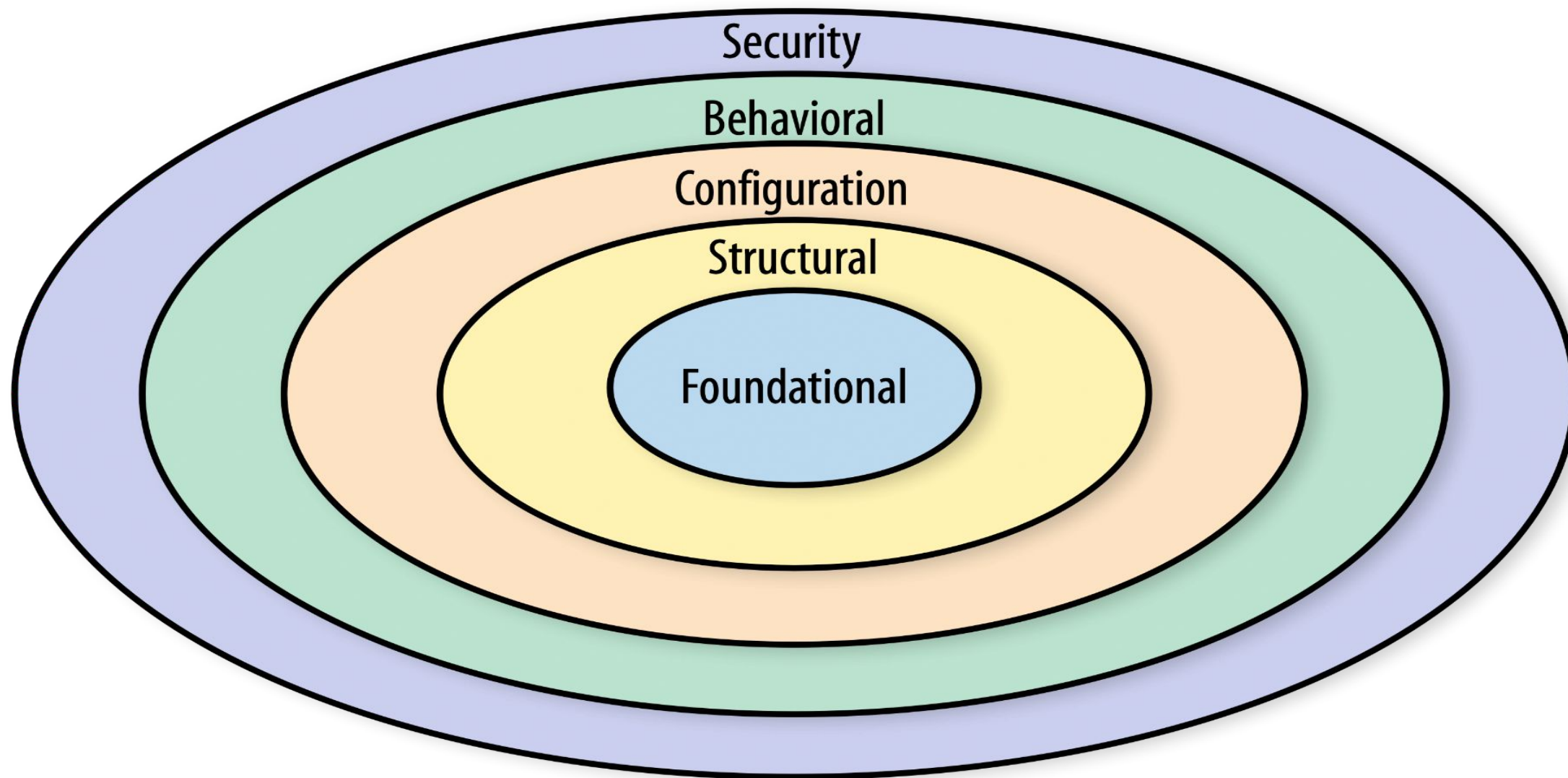




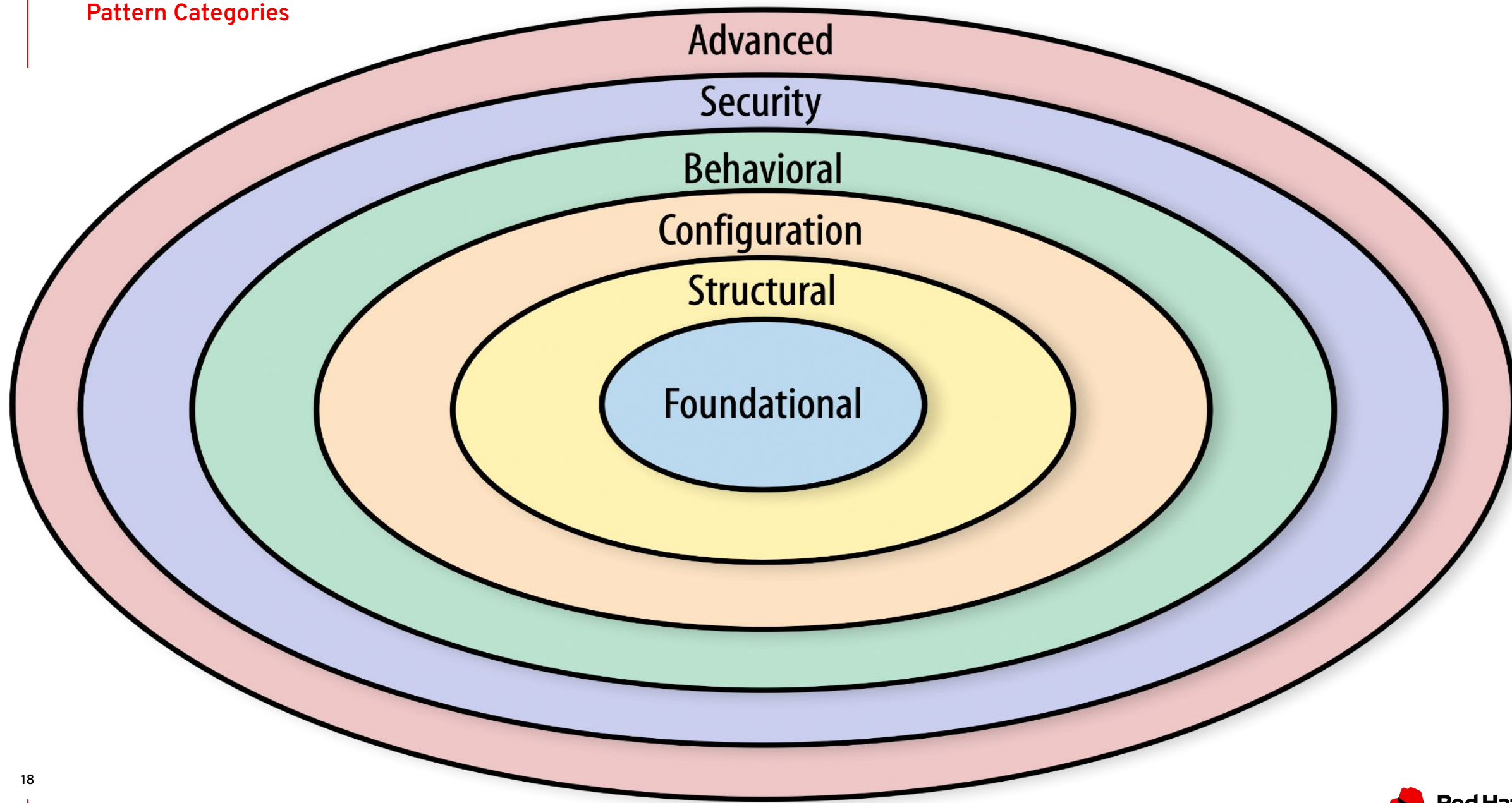


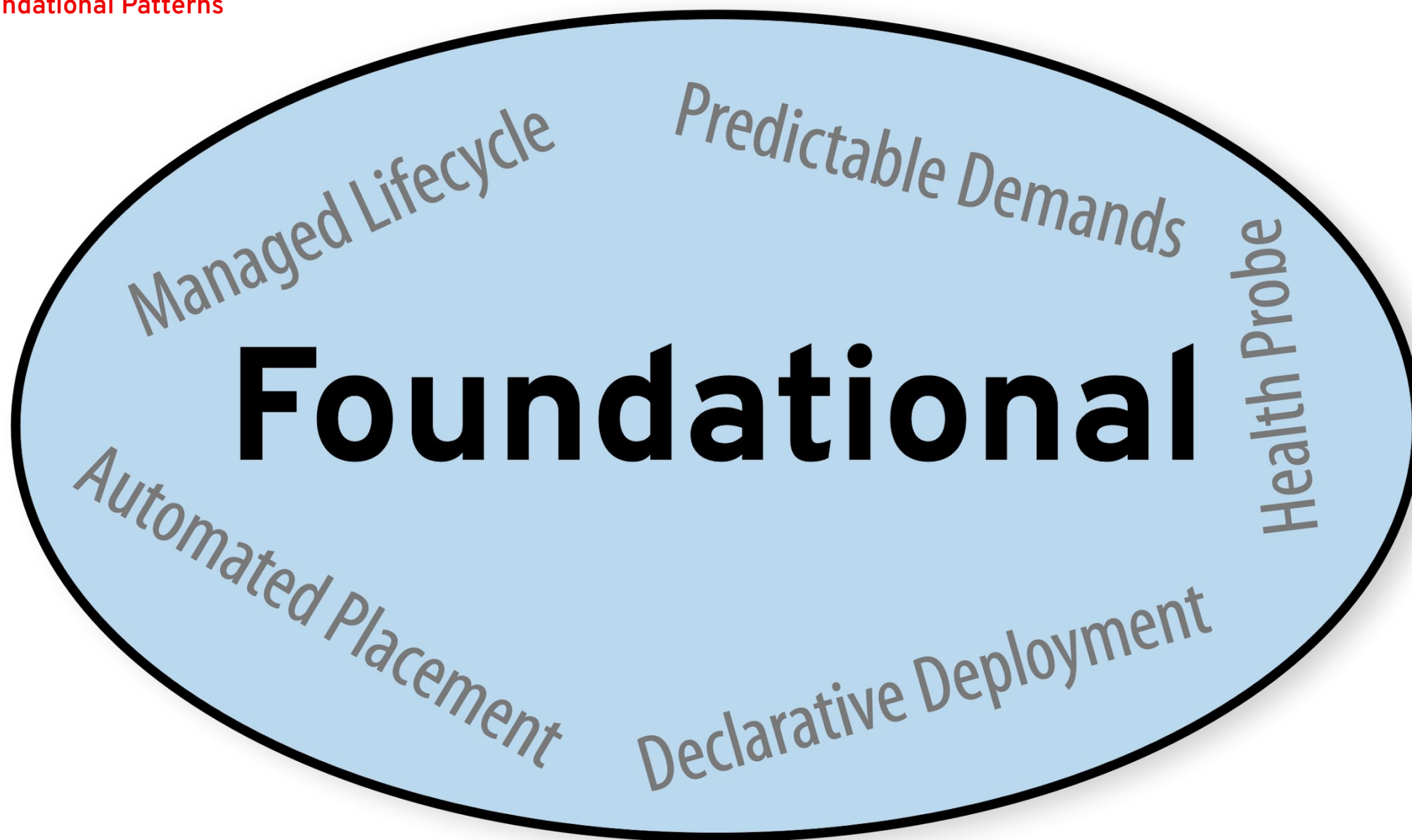






Pattern Categories







Predictable Demands

How to declare application requirements

Application Requirements

- Declared requirements help in
 - Scheduling decisions
 - Capacity planning
 - Matching infrastructure services
- Hard runtime dependencies
 - Persistent Volumes
 - Host ports
 - Dependencies on ConfigMaps and Secrets

Resource Profile

```
apiVersion: v1
kind: Pod
metadata:
  name: http-server
spec:
  containers:
  - image: nginx
    name: nginx
    resources:
      requests:
        cpu: 200m
        memory: 100Mi
      limits:
        cpu: 300m
        memory: 200Mi
```

Quality-of-Service Classes

- **Best Effort**
 - No requests or limits
- **Burstable**
 - requests < limits
- **Guaranteed**
 - requests == limits

Recommendations

For **memory**, set requests **equal** to limits

For **cpu**, set requests, but **no limits**

References:

- [For the love of god, stop using CPU limits on Kubernetes](#)
- [What everyone should know about Kubernetes memory limits, OOMKilled pods, and pizza parties](#)

Thomas Peitz · May 29, 2019
@tpeitz_dus · [Follow](#)

We have reduced 75 percentile response time over all apps from 150ms to 90ms after disabling CFS quota (CPU limits) on one of our [#kubernetes](#) cluster - [#KubeCon](#) learning by [@try_except_](#)



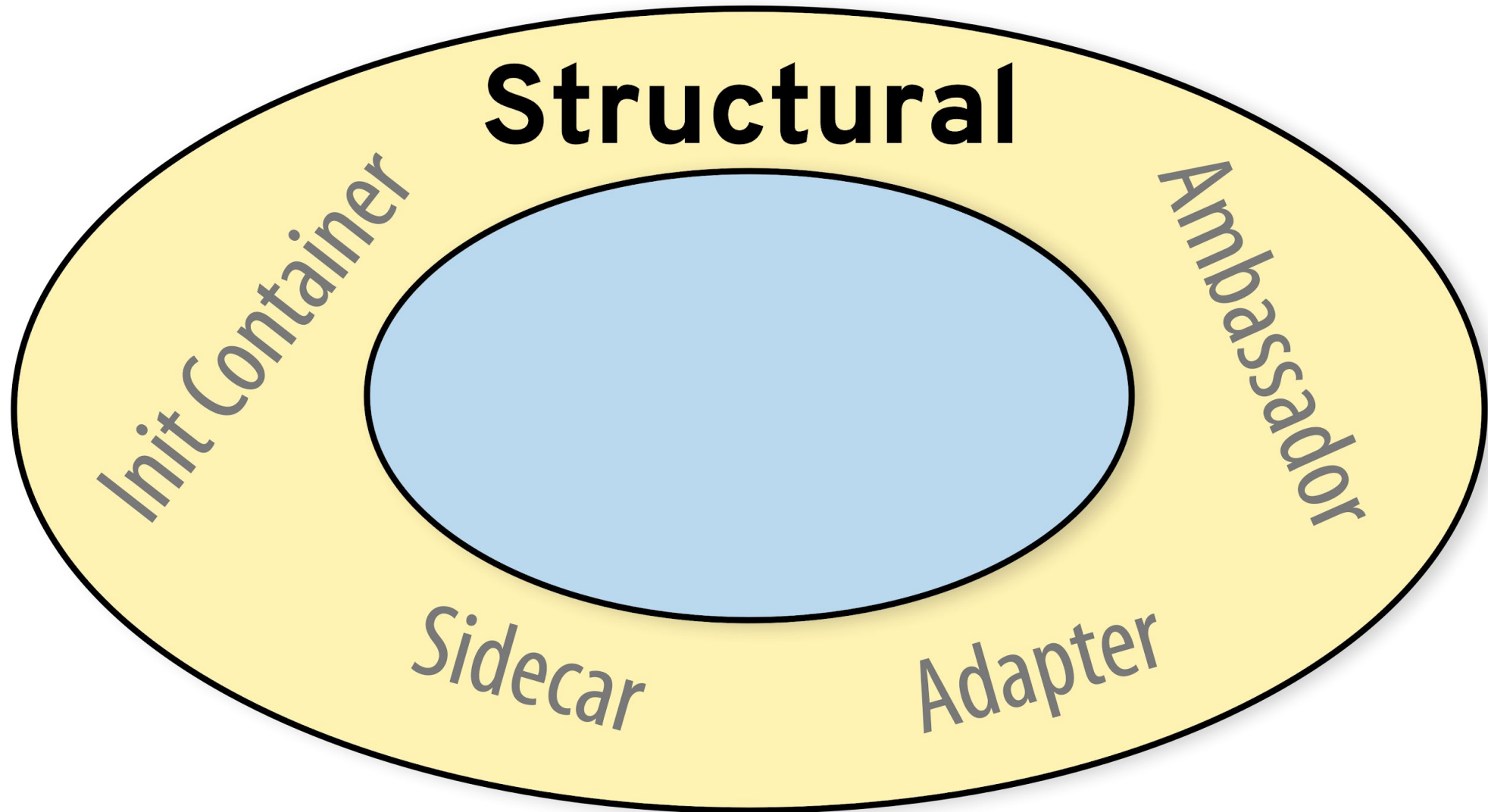
Tim Hockin (thockin.yaml)
@thockin · [Follow](#)

This is why I always advise:

- 1) Always set memory limit == request
- 2) Never set CPU limit

(for locally adjusted values of "always" and "never")

10:24 PM · May 30, 2019



A photograph of a sidecar motorcycle race on a muddy track. In the foreground, a rider in a maroon and white suit is on the motorcycle, while a passenger in a black suit is in the sidecar. They are both covered in mud. In the background, two other riders are visible. The track is muddy and there is a grassy bank on the left.

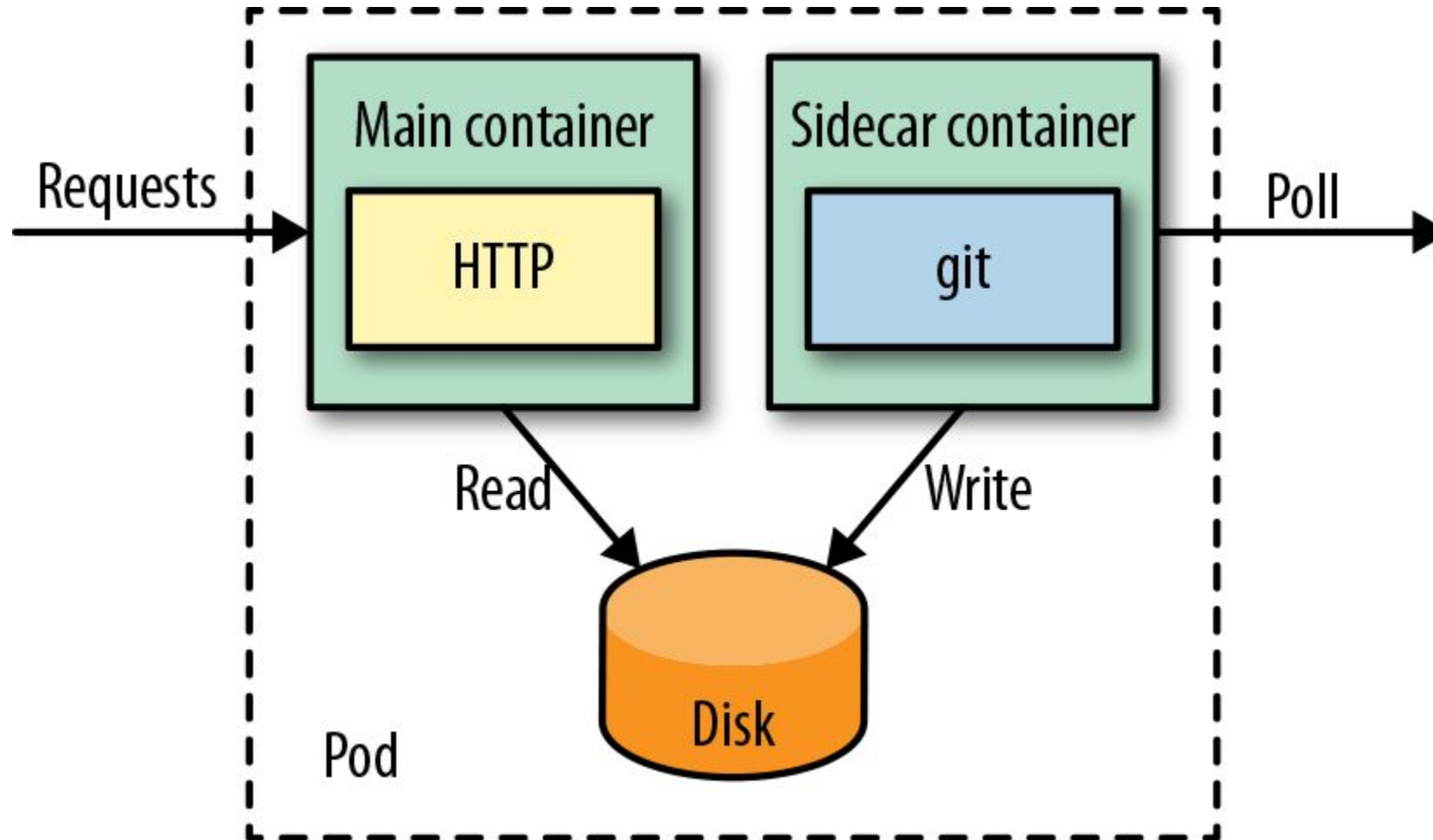
Sidecar

How to enhance the functionality of an application without changing it

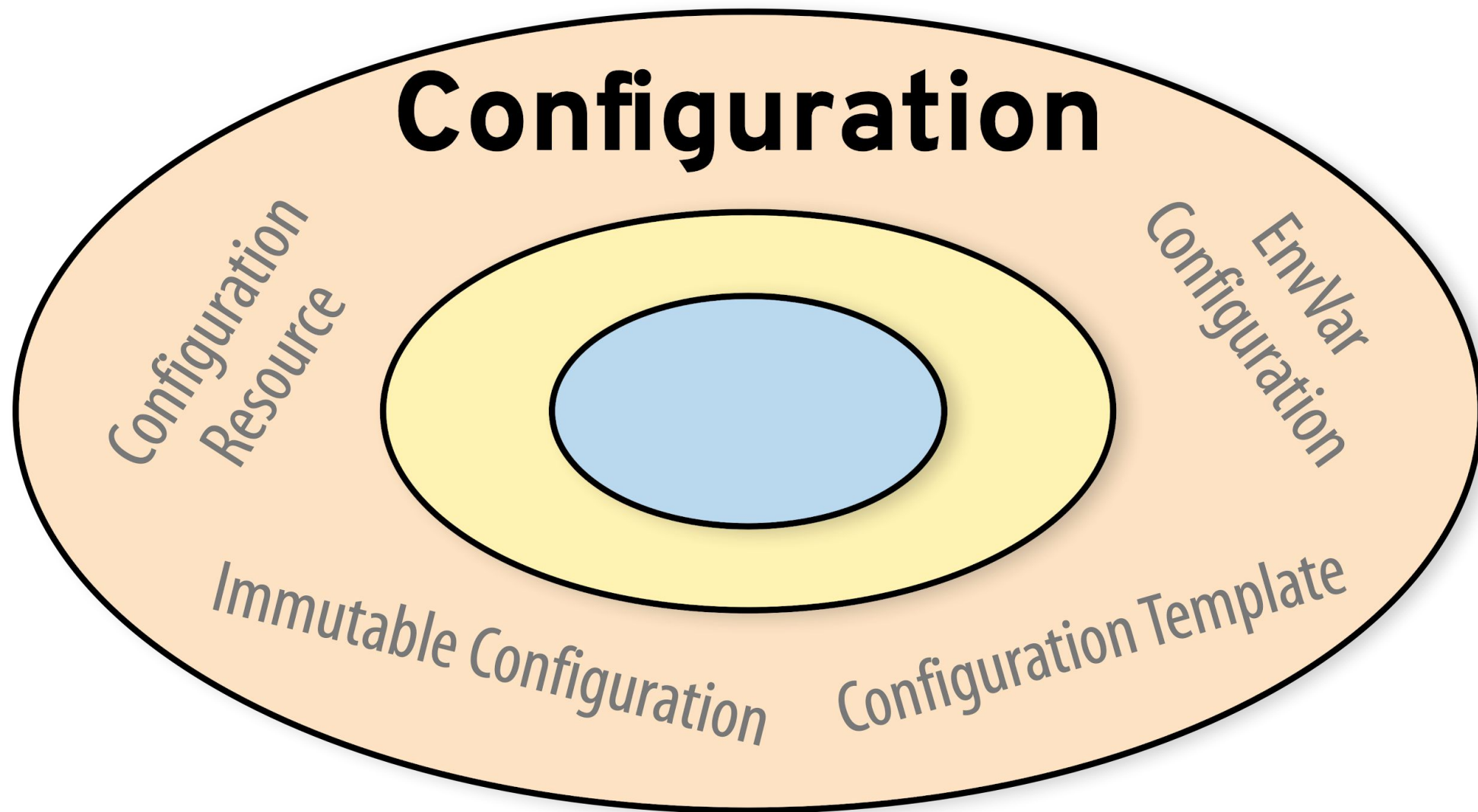
Sidecar

- Runtime collaboration of containers
- Connected via shared resources:
 - Network
 - Volumes
- Similar what AOP is for programming
- Separation of concerns

Sidecar



Demo





Configuration Template

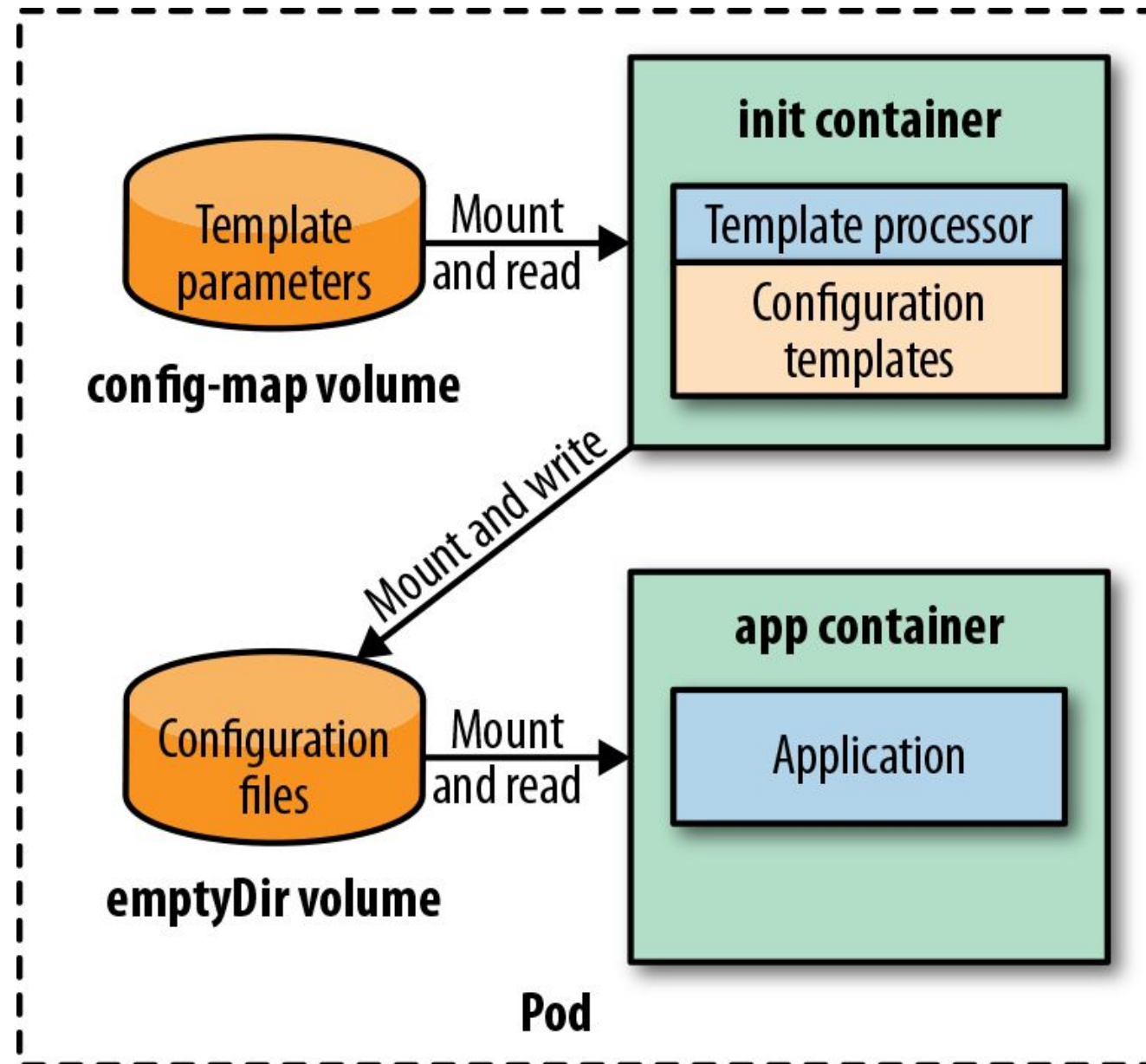


How to manage large and complex similar configuration data

Preparing Configuration during Startup

- Init Container ...
 - ... contains a template processor
 - ... holds the configuration template
 - ... picks up template parameter from a ConfigMap
 - ... stores final configuration on a shared volume
- Main Container
 - ... accesses created configuration from shared volume

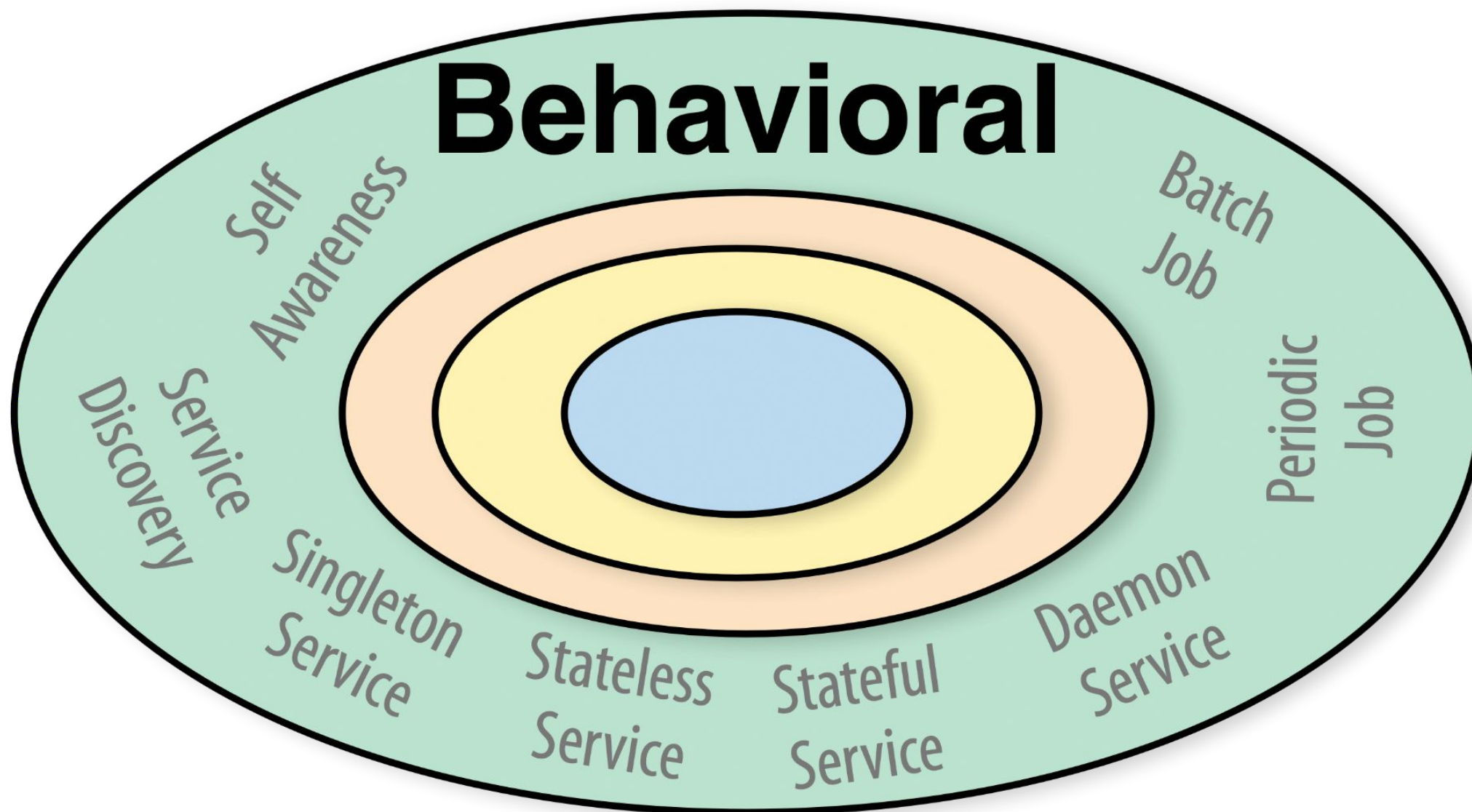
Configuration Template



Configuration Template

```
apiVersion: extensions/v1beta1
kind: Deployment
metadata:
  name: wildfly-cm-template
spec:
  replicas: 1
  template:
    spec:
      initContainers:
        - image: k8spatterns/config-init
          name: init
          volumeMounts:
            - mountPath: "/params"
              name: wildfly-parameters
            - mountPath: "/out"
              name: wildfly-config
```

```
containers:
- image: jboss/wildfly:10.1.0.Final
  name: server
  volumeMounts:
    - mountPath: "/config"
      name: wildfly-config
volumes:
- name: wildfly-parameters
  configMap:
    name: wildfly-params-cm
- name: wildfly-config
  emptyDir: {}
```

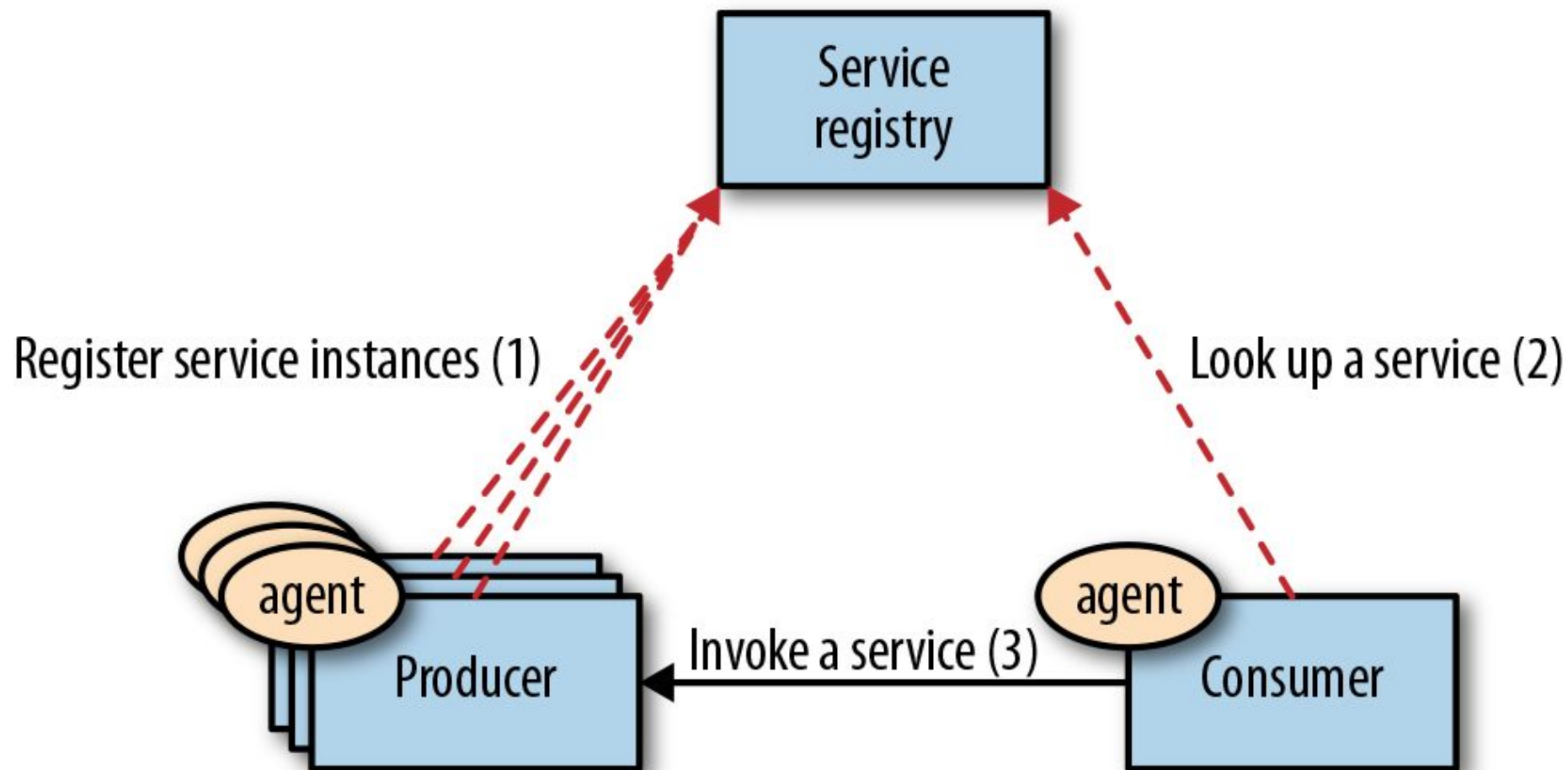





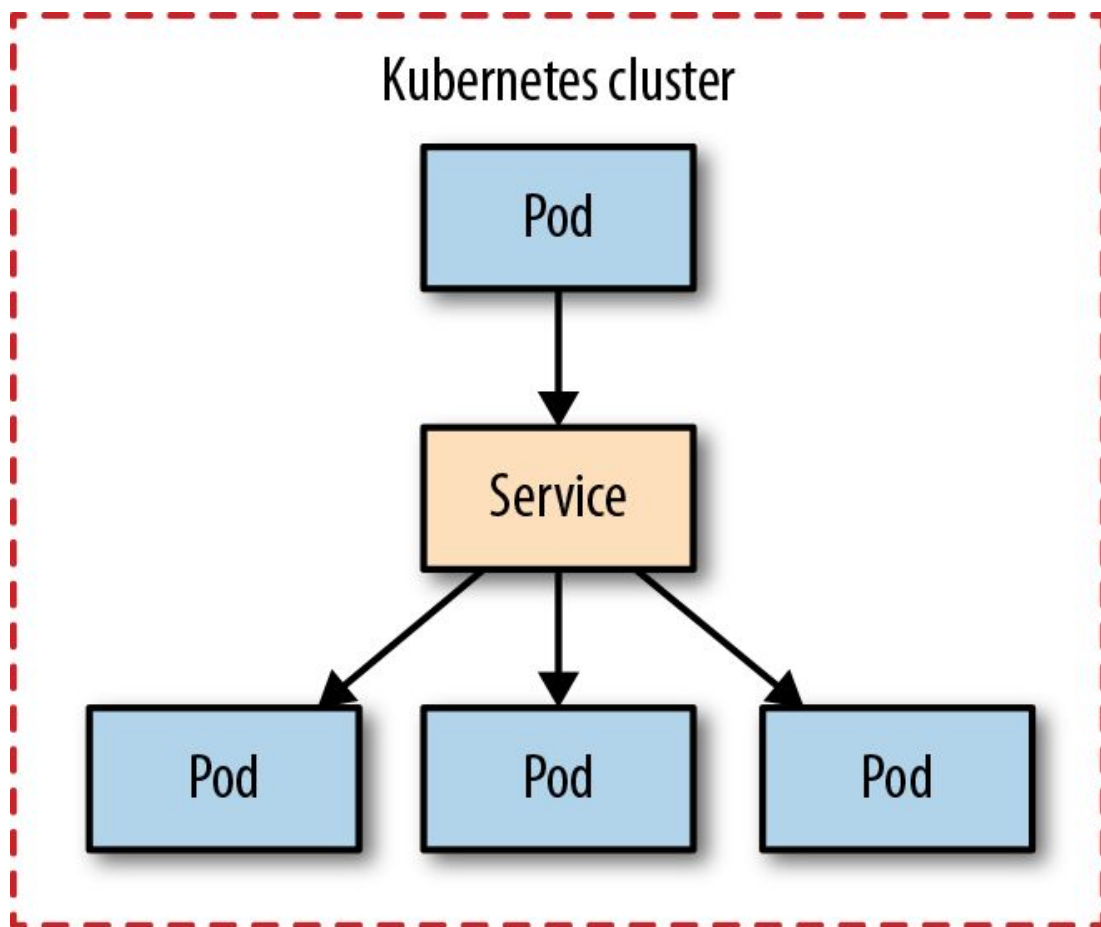
Service Discovery

How to discover and use services

Client-side Service Discovery (non Kubernetes)



Internal Service Discovery

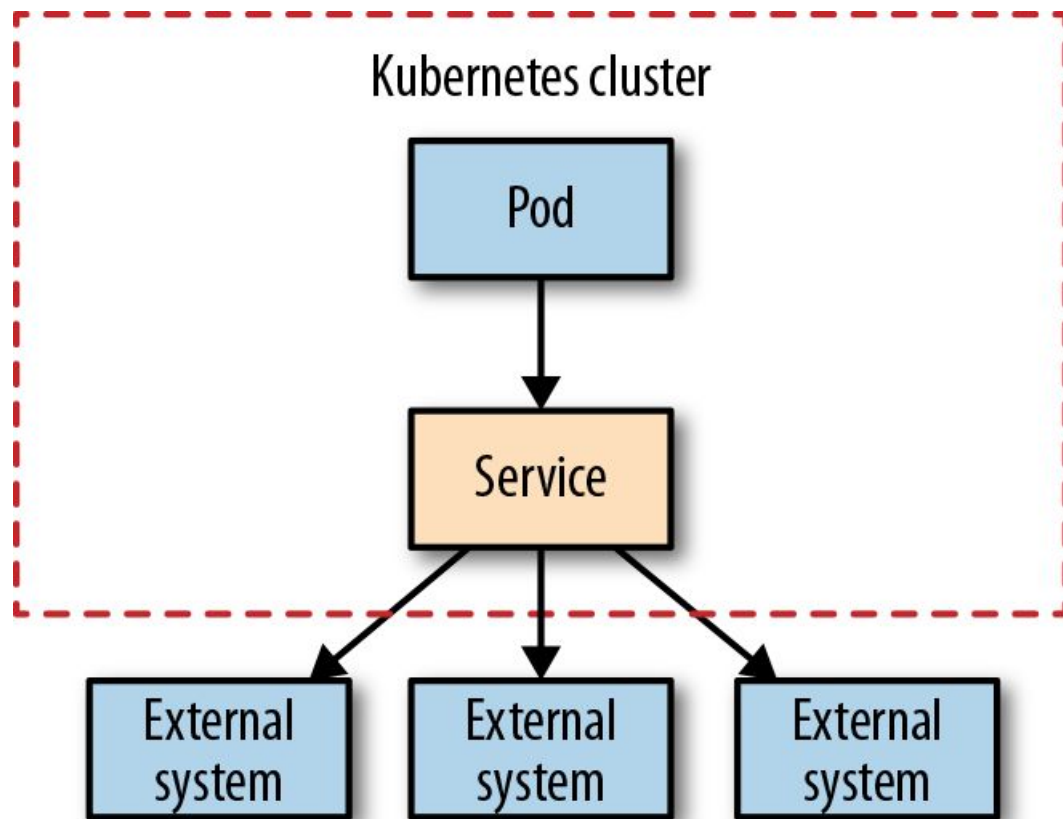


- Discovery through DNS lookups
- Pods picked by label selector
- Multiple ports per Service
- Session affinity on IP address possible
- Successful readiness probes required for routing
- Virtual IP address for each Service

Service

```
apiVersion: v1
kind: Service
metadata:
  name: random-generator
spec:
  selector:
    app: random-generator
  ports:
    - port: 8080
      protocol: TCP
      targetPort: 8080
```


Manual Service Discovery

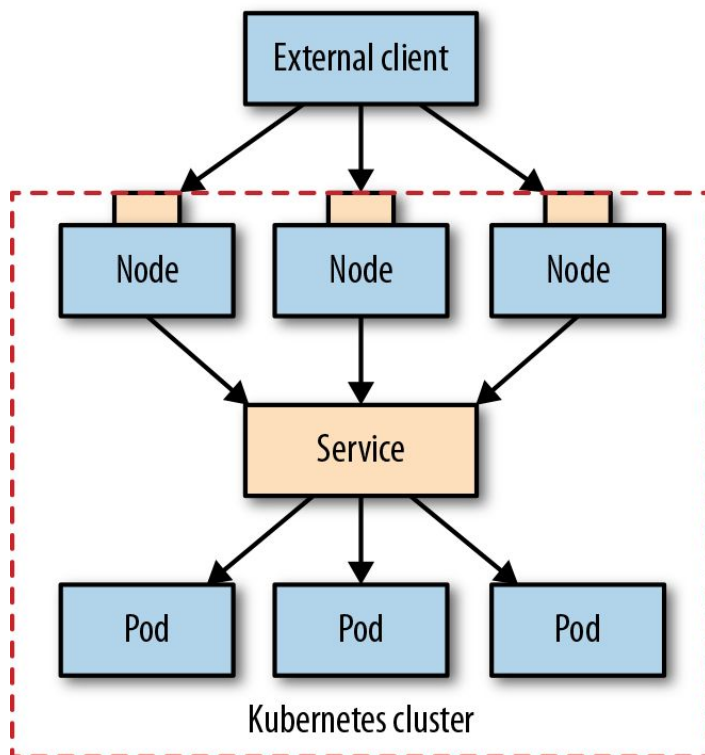


- Service without selector
- Manually creating Endpoint resource with the same name as the Service
- Service of type **ExternalName** map are registered as DNS CNAMEs

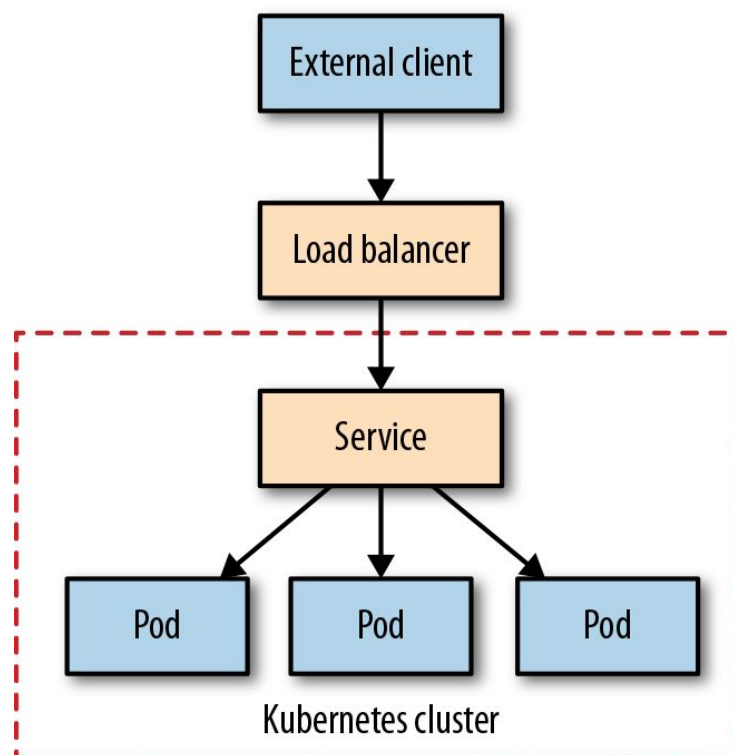
Endpoints

```
apiVersion: v1
kind: Endpoints
metadata:
  name: external-service
subsets:
- addresses:
  - ip: 1.1.1.1
  - ip: 2.2.2.2
  ports:
  - port: 8080
```

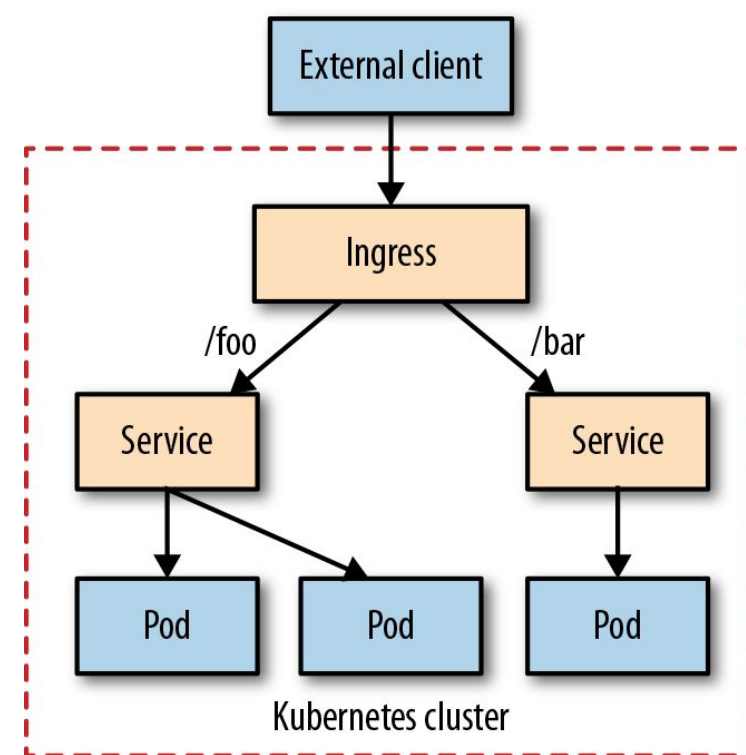
Node Port



Load Balancer



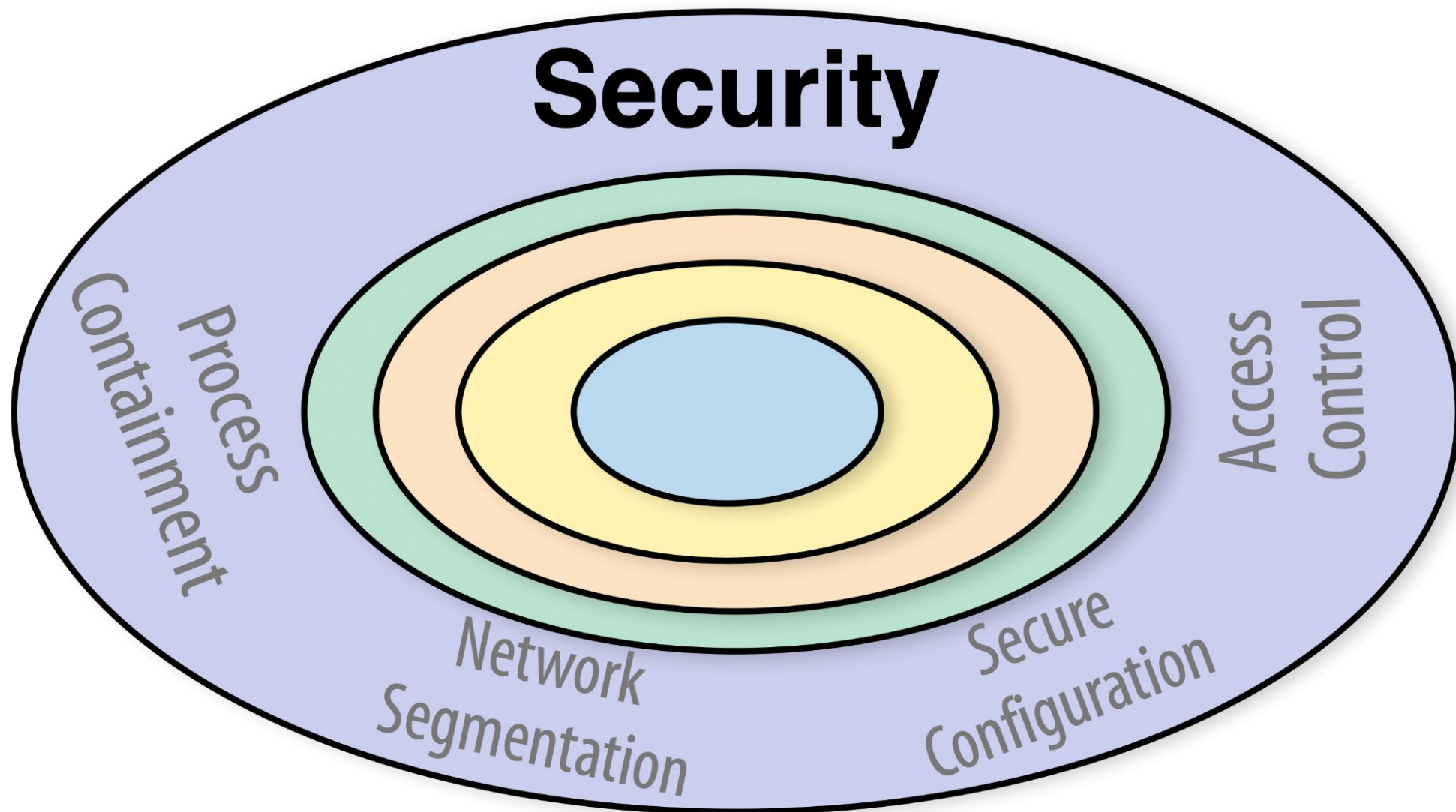
Ingress

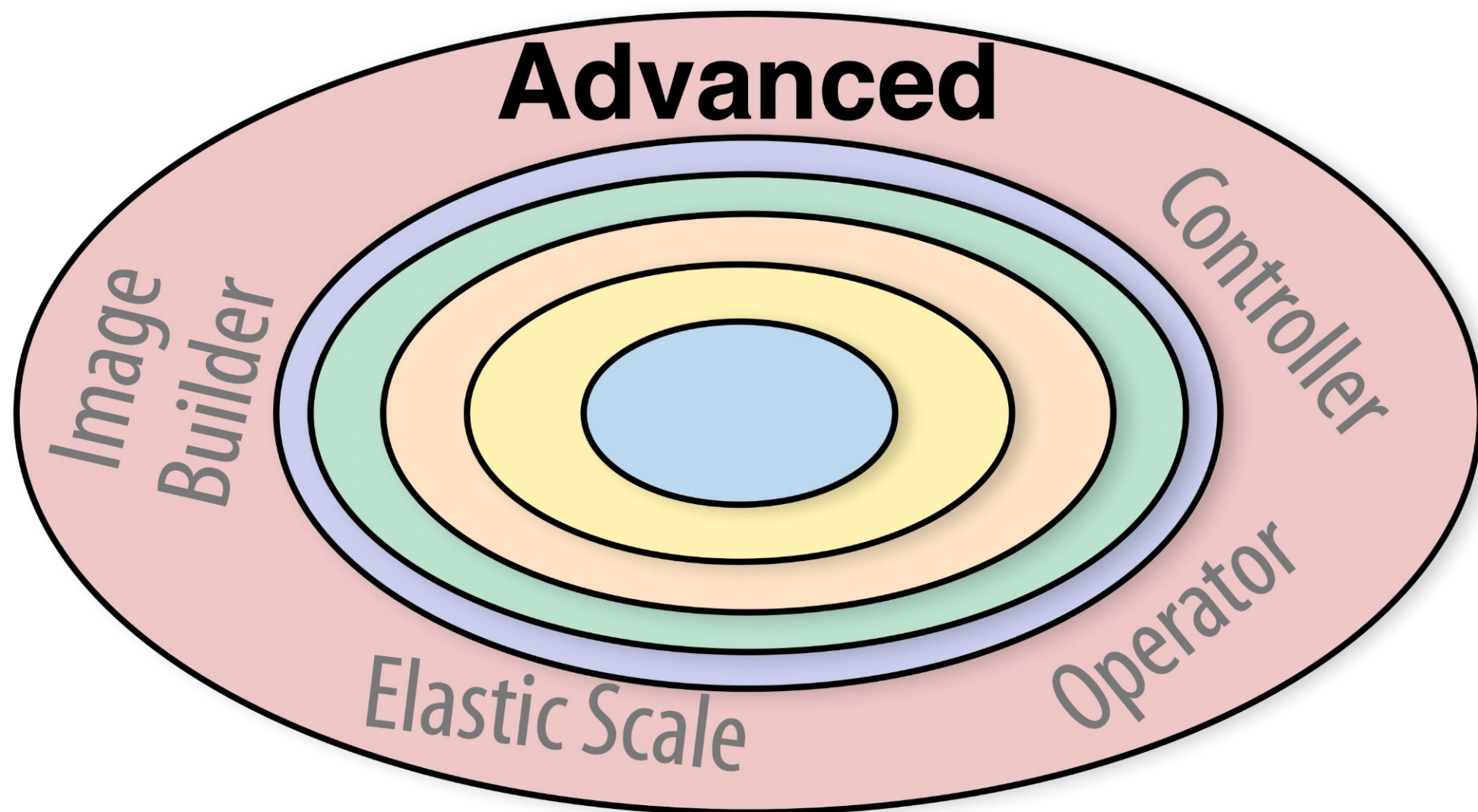


Ingress

```
apiVersion: extensions/v1beta1
kind: Ingress
metadata:
  name: random-generator
spec:
  rules:
  - http:
    paths:
    - path: /
      backend:
        serviceName: random-generator
        servicePort: 8080
    - path: /cluster-status
      backend:
        serviceName: cluster-status
        servicePort: 80
```

Name	Configuration	Client type	Summary
ClusterIP	type: ClusterIP .spec.selector	Internal	The most common internal discovery mechanism
Manual IP	type: ClusterIP kind: Endpoints	Internal	External IP discovery
Manual FQDN	type: ExternalName .spec.externalName	Internal	External FQDN discovery
Headless Service	type: ClusterIP .spec.clusterIP: None	Internal	DNS-based discovery without a virtual IP
NodePort	type: NodePort	External	Preferred for non-HTTP traffic
LoadBalancer	type: LoadBalancer	External	Requires supporting cloud infrastructure
Ingress	kind: Ingress	External	L7/HTTP-based smart routing mechanism





Controller



How to get from the
current state to a
declared target state

State Reconciliation

- Kubernetes as distributed state manager
- Make the **actual** state more like the declared **target** state.



Observe - Discover the actual state

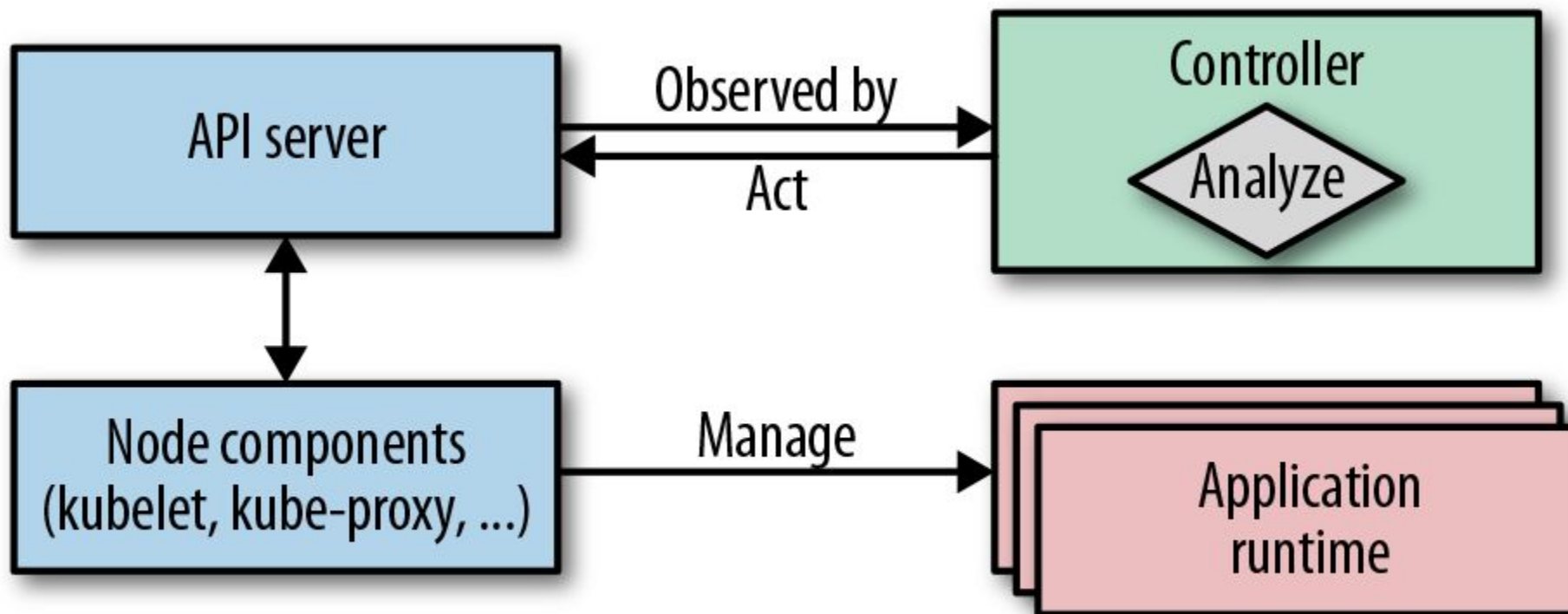


Analyze - Determine difference to target state



Act- Perform actions to drive the actual to the desired state

Observe - Analyze - Act



Common Triggers

- Labels
 - Indexed by backend
 - Suitable for selector-like functionality
 - Limitation on charset for names and values
- Annotations
 - No syntax restrictions
 - Not indexed
- ConfigMaps
 - Good for complex structured state declarations
 - Simple alternative to CustomResourceDefinitions

ConfigMap Watch Controller

```
namespace=${WATCH_NAMESPACE:-default}
base=http://localhost:8001
ns=namespaces/$namespace

curl -N -s $base/api/v1/${ns}/configmaps?watch=true | \
while read -r event
do
    type=$(echo "$event" | jq -r '.type')
    config_map=$(echo "$event" | jq -r '.object.metadata.name')
    annotations=$(echo "$event" | jq -r '.object.metadata.annotations')

    if [ $type = "MODIFIED" ]; then
        # Restart Pods using this ConfigMap
        # ...
    fi
done
```



Operator

How to encapsulate
operational knowledge
into executable software

Definition

“““ An **operator** is a Kubernetes **controller** that understands two domains: Kubernetes and *something else*. By combining knowledge of both areas, it can **automate** tasks that usually require a human operator that understands both domains.

Jimmy Zelinskie

<http://bit.ly/2Fjlx1h>

Technical:

Operator = Controller + CustomResourceDefinition

CustomResourceDefinition

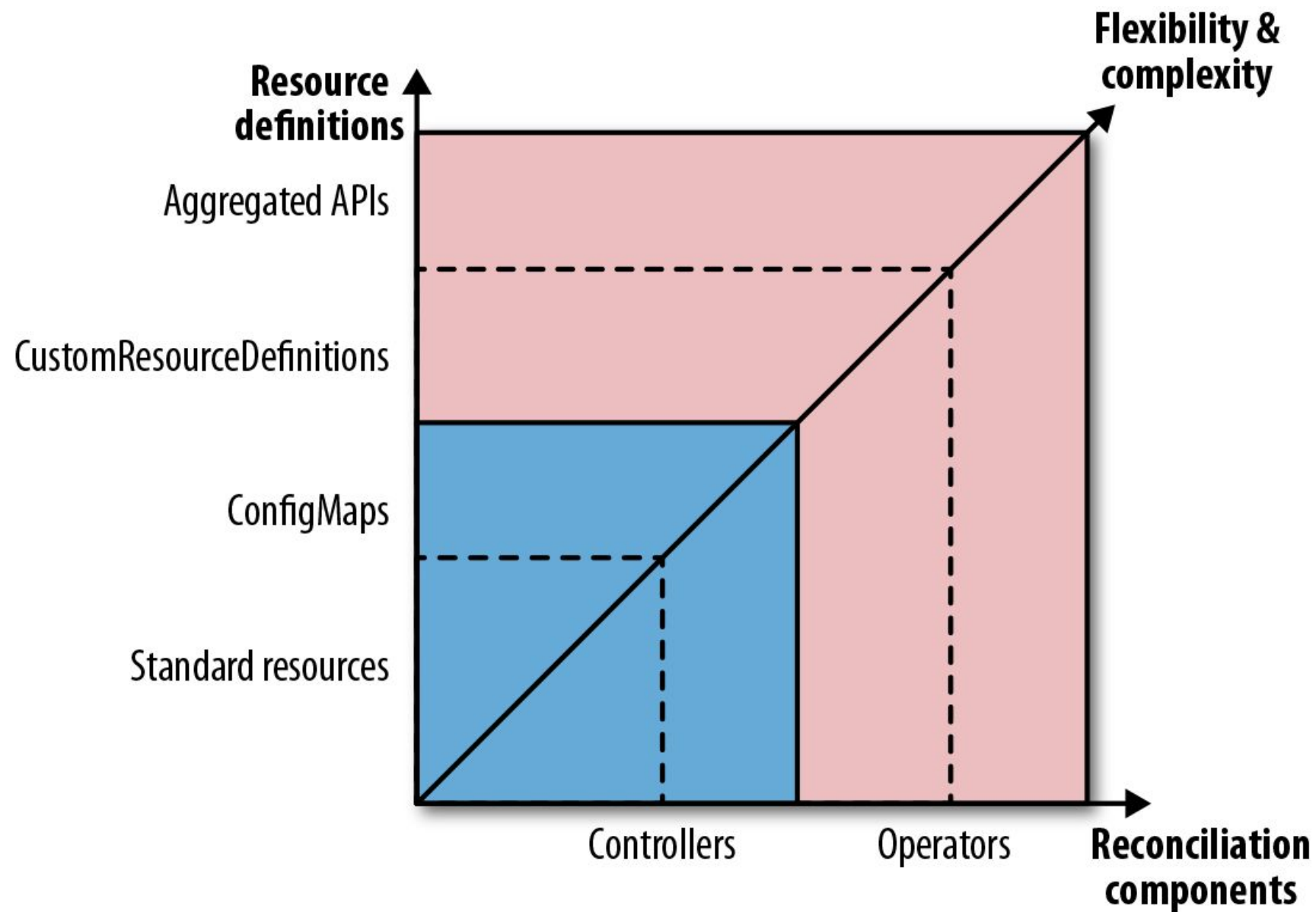
Custom resource is modelling a custom domain and managed through the Kubernetes API

```
apiVersion: apiextensions.k8s.io/v1beta1
kind: CustomResourceDefinition
metadata:
  name: configwatchers.k8spatterns.io
spec:
  scope: Namespaced
  group: k8spatterns.io
  version: v1
  names:
    kind: ConfigWatcher
    plural: configwatchers
  validation:
    openAPIV3Schema:
      ...
```

Custom Resource

Operator

```
kind: ConfigWatcher
apiVersion: k8spatterns.io/v1
metadata:
  name: webapp-config-watcher
spec:
  configMap: webapp-config
  podSelector:
    app: webapp
```



CRD Classification

- Installation CRDs
 - Installing and operating applications
 - Backup and Restore
 - Monitoring and self-healing
 - Example: Prometheus for installing Prometheus & components
- Application CRDs
 - Application specific domain concepts
 - Example: ServiceMonitor for registering Kubernetes service to be scraped by Prometheus

Operator Hub

The screenshot shows the OperatorHub.io website interface. The header includes the site name, a search bar, and a 'Contribute' link. The main heading is 'Welcome to OperatorHub.io' with a subtext: 'OperatorHub.io is a new home for the Kubernetes community to share Operators. Find an existing Operator or list your own today.'

The main content area displays a list of operators, categorized by 'PROVIDER' and 'CAPABILITY LEVEL'. The 'PROVIDER' section shows a list of providers with checkboxes and counts. The 'CAPABILITY LEVEL' section shows a list of capability levels with checkboxes and counts.

PROVIDER

- ☐ Amazon Web Services (1)
- ☐ CNCF (1)
- ☐ Couchbase (1)
- ☐ Crunchy Data Solutions (1)
- ☐ Dynatrace (1)
- ☒ Jaeger (1)
- ☒ MongoDB (1)
- ☐ Percona (1)
- ☐ PlanetScale (1)
- ☒ Red Hat (3)
- ☐ Redis Labs (1)
- [Show less](#)

CAPABILITY LEVEL

- ☐ Basic Install (4)
- ☐ Seamless Upgrades (3)
- ☐ Full Lifecycle (6)

5 ITEMS

VIEW **SORT** A-Z

Operators:

- Jaeger Tracing**
provided by Jaeger
Provides tracing, monitoring and troubleshooting microservices-based
- Kubernetes Federation**
provided by Red Hat
Gain Hybrid Cloud capabilities between your clusters with Kubernetes Federation.
- MongoDB**
provided by MongoDB, Inc
The MongoDB Enterprise Kubernetes Operator enables easy deploys of MongoDB
- Prometheus Operator**
provided by Red Hat
The Prometheus Operator for Kubernetes provides easy monitoring definitions for
- Strimzi Kafka**
provided by Red Hat
Run an Apache Kafka cluster, including Kafka Connect, ZooKeeper and more.

Operator Development

- Operator can be implemented in any language
- Frameworks:
 - Operator Framework (Golang, Helm, Ansible, Java)
<https://github.com/operator-framework>
 - Kubebuilder (Golang)
<https://github.com/kubernetes-sigs/kubebuilder>
 - Metacontroller (Language agnostic)
<https://metacontroller.app/>

Thank you



<https://k8spatterns.com>



@ro14nd@hachyderm.io



www.linkedin.com/in/ro14nd

Picture Credits

<https://www.pexels.com/photo/brown-and-black-pattern-2158386/>
<https://pixabay.com/photos/ship-helm-sunset-cutter-coast-guard-759954/>
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